LieRing Programmer Manual

modules, scripts, accepted design choices

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1 Programmer Manual

Physics is in the Beam Physics Manual. This page lists entry points of the Julia package.

The object of the ring is the Lie product of Dragt–Finn factors. That is the same cut as MaryLie: concatenate in the algebra, analyse the product, track a point through it. Linear optics is the closed form of $\exp(: f_2 :)$, stored as the matrix R. It is not a second theory of the lattice.

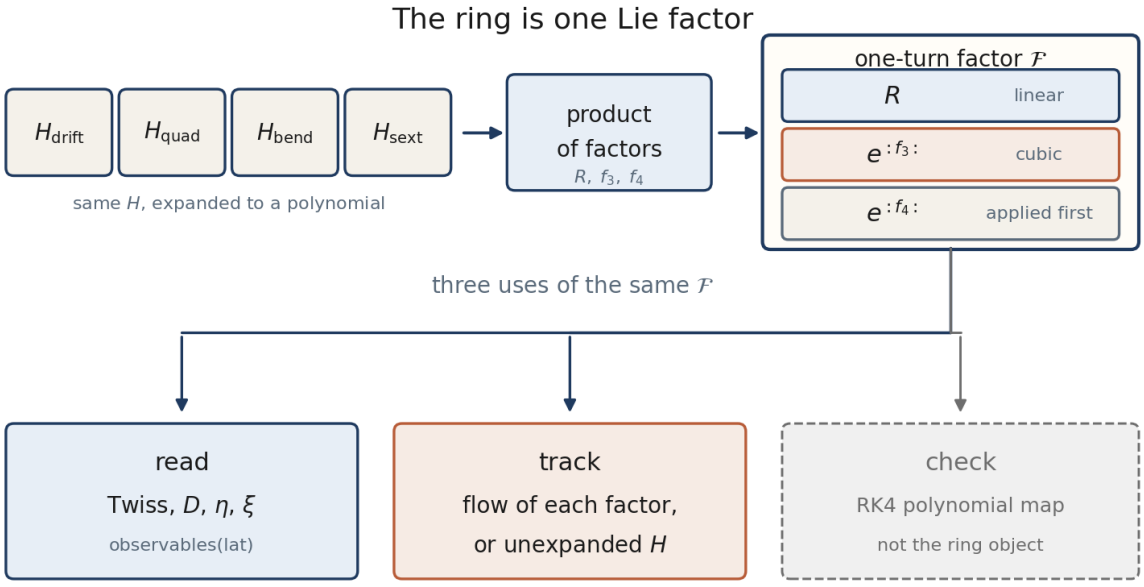


Figure 1: The ring is one Lie factor: Hamiltonians are concatenated in the algebra; the product is read, tracked, or checked.

`observables(lat)` builds `ring_lie_factor` once and reads Twiss, η and ξ from that factor. `ring_taylor_map` is an RK4 polynomial and is not that object. `ring_matrix` remains a fast order-1 builder for plots along s.

1.1 Layout

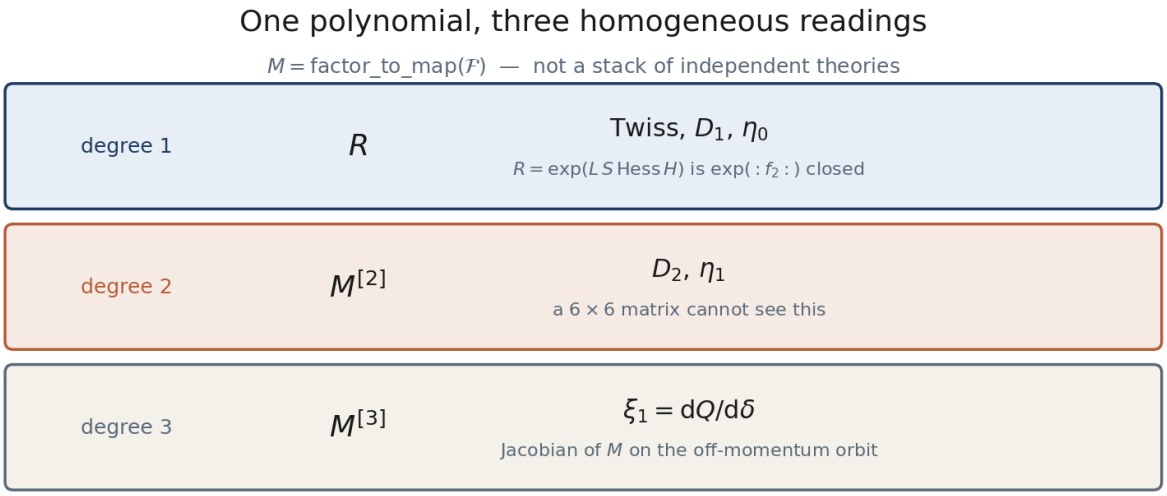


Figure 2: One polynomial, three homogeneous readings: degree 1 gives Twiss and η_0; degree 2 gives D_2 and η_1; degree 3 gives ξ_1.

path	**role**
`src/Constants.jl`	CODATA deuteron, reference particle, $v(\delta)$, $e(\delta)$, $z \leftrightarrow u$
`src/Jets.jl`	polynomial space, degree 4 in Hamiltonians, degree 3 in maps
`src/Elements.jl`	element types and lattice
`src/Hamiltonians.jl`	element generator H as a polynomial
`src/LieOperators.jl`	Poisson bracket, Lie series, Dragt–Finn, product of factors to degree 4
`src/OptiMParser.jl`	`.opt` reader (mass from the particle, not the file header)
`src/LinearOptics.jl`	$R = \exp(LS\mathrm{Hess}\, H)$, slice Twiss along s
`src/Symplectic.jl`	$J^{\mathsf{T}} S J - S$
`src/Tracking.jl`	flow of each factor, thin RF kick, product of the ring
`src/MapAnalysis.jl`	closed orbit, D_2, η, chromaticity, `observables`
`src/TaylorMap.jl`	independent DA map (RK4 in polynomials) — a check
`src/NormalForm.jl`	one-turn factor with thin RF; homological operator on generators; torus average of δ
`src/SecondOrder.jl`	averaged longitudinal Hamiltonian, BLonD (dt, dE) bridge
`src/ExactIntegrator.jl`	unexpanded H, finite cavity, renormalise
`src/Acceleration.jl`	ramp, tune fit
`src/Plotting.jl`	figures
`src/Optimizers.jl`	optional sextupole response

Entry: `include("src/LieRing.jl"); using .LieRing`.

The one-turn object used for the normal form is `one_turn_factor_rf`, the product of `ring_lie_factor` with the thin-kick factor. Optics of the magnetic product without RF is `observables(lat)`. `ring_taylor_map` composes truncated polynomials and is not that object.

1.2 Scripts

```
julia --project=. scripts/01_linear_optics.jl
julia --project=. scripts/02_lie_maps.jl
```

```
julia --project=. scripts/03_second_order.jl
julia --project=. scripts/04_optional_sextupole_optimization.jl
julia --project=. scripts/05_rf_acceleration.jl
julia --project=. scripts/06_coordinates.jl
julia --project=. scripts/07_blond_longitudinal.jl   # external drift formulae, not used in t
julia --project=. scripts/08_lie_product.jl          # BCH product of two factors
julia --project=. scripts/09_lie_normal_form.jl      # one-turn factor and torus average
```

The equilibrium-momentum study lives in `C:\GIT_REPOS\SCT_2\Julia\equilibrium_energy` (`scripts/run_v0` and the research note). It depends on this package via `Pkg.develop`.

1.3 Accepted design choices

- Injection kinetic energy 10 MeV; final 270 MeV from the lattice header. - Stationary study: thin RF, $V = 0.2$ MV, $h = 5$, $\varphi_s = \pi$. - Ramp demo: finite cavity $L = 1$ m, $\varphi_s = 150°$. - Mass = CODATA deuteron 1875.61294257 MeV (OptiM `Mass=2*938.272` ignored). - Lattice sequence used once (file `Number of periods=2` is an error; bend is 360 deg). - Closed $D(0) = 0.011$ m (OptiM initial 0 is not the closed solution). - Default sextupole strengths are zero.

1.4 Docs build

```
julia --project=docs docs/make.jl
julia docs/make_pdf.jl
julia docs/make_pdf_programmer.jl
```